

RHEA SUDHEER

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SUMMARY

Software engineer with an M.S. in Computer Science from Texas A&M and hands-on experience building backend services, SQL-backed platforms, REST/API integrations, concurrent systems, and low-level software. Strong in Python, C++, Java, SQL/MySQL, object-oriented design, data structures & algorithms, debugging, testing, and scalable system design. Proven ability to translate ambiguous requirements into reliable features, investigate application/database failures, and deliver maintainable systems through Git, CI/CD, code review, and cross-functional collaboration.

EDUCATION

Texas A&M University — M.S. Computer Science, GPA 3.83/4.00 | College Station, TX

Aug 2024 – May 2026

PES University — B.Tech. Computer Science, GPA 9.1/10 | Bangalore, India

Dec 2020 – May 2024

Relevant Coursework: Analysis of Algorithms, Operating Systems, Information Storage & Retrieval, Machine Learning, Computer/Network Security

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL, PHP, JavaScript, Bash

Software Engineering: Object-Oriented Design, Data Structures & Algorithms, REST/API Development & Integration, SDLC, Production Debugging, Unit Testing, Code Review

Backend / Data: Laravel, Flask, MySQL, Relational Schemas, ETL, Data Validation, SQL Analytics, Spark, Kafka

Systems: Linux, Distributed Systems, Multithreading, Synchronization, Virtual Memory, Scheduling, Thread-Safe Queues, Backpressure

Cloud / DevOps / Tools: Git, CI/CD, Docker, Kubernetes, AWS, GCP, Jenkins, GDB/LLDB, CMake

EXPERIENCE

Public Policy Research Institute, Texas A&M University — Student Research Assistant

Jan 2025 – Present

- Engineered and maintained a PHP/Laravel + MySQL research platform supporting 100,000+ public records; built backend services, relational schemas, REST/API integrations, data-collection workflows, dashboards, and reporting features.
- Developed API-backed ETL pipelines and reusable SQL transformations for cleaning, joining, aggregating, and validating multi-source data; automated checks for malformed, incomplete, duplicated, and inconsistent records before downstream use.
- Diagnosed application, database, and data-quality failures using logs, SQL investigation, targeted tests, and profiling; converted recurring failure modes into fixes, stronger validation logic, and regression protection.
- Improved maintainability through unit testing, CI checks, code review, Git-based workflows, documentation, and release support; partnered with researchers and developers to translate evolving requirements into technical specifications, schema changes, and iterative releases.
- Built stakeholder-facing dashboards and recurring reports, prioritized competing technical requests, and coordinated changes so application behavior, database structure, and reporting outputs remained aligned as requirements evolved.

Ernst & Young — Technology Consulting Intern | Bangalore, India

Jan 2024 – Jun 2024

- Evaluated enterprise Oracle/SAP systems for access management, change management, disaster recovery, security, and operational reliability; traced production changes through approvals, implementation evidence, and governance workflows.
- Investigated control exceptions with client technology teams, performed root-cause analysis, and translated technical issues into traceable remediation actions, repeatable reporting workflows, and audit-ready documentation.

Thales — Artificial Intelligence Intern | Bangalore, India

Jun 2022 – Aug 2022

- Developed a real-time MobileNetV2 + SSD pedestrian-detection pipeline; profiled latency, accuracy, missed/false detections, and deployment constraints, and used SHAP/LIME/saliency analysis as debugging and model-validation tools.
- Balanced model quality, response-time constraints, interpretability, and reliability instead of optimizing offline accuracy alone; documented experimental observations and communicated trade-offs to technical stakeholders.

PROJECTS

Custom x86 Operating System | C/C++, x86, QEMU/Bochs, GDB/LLDB

- Implemented demand-paged virtual memory with page tables, page-fault handling, frame allocation, and TLB-aware management; built timer-driven preemptive scheduling with ready queues and safe thread lifecycle management.
- Implemented spinlocks/mutexes, an inode-based file system, buffered I/O, and interrupt-driven non-blocking disk access using request queues; debugged memory, scheduling, synchronization, and device-I/O failures at the hardware/software boundary.

Adaptive Multithreaded Job Scheduler with Backpressure | C++, *multithreading, mutexes, condition variables*

- Built a concurrent scheduler supporting FIFO and priority policies with thread-safe queues, explicit locking boundaries, and condition-variable-based worker coordination; separated scheduling policy from worker execution.
- Added backpressure and runtime metrics for queue depth, job wait time, and worker utilization to analyze overload behavior and identify queueing, capacity, and scheduling bottlenecks.

Hybrid Self-RAG with External Knowledge Validation | Python, FAISS, BM25, LangChain, APIs

- Built a multi-stage retrieval and verification system integrating external APIs, claim extraction, semantic scoring, FAISS/BM25 fallback retrieval, and evidence-conditioned regeneration; improved factual accuracy from 81% to 89%, reduced hallucination rate from 19% to 11%, and increased average FactScore from 0.65 to 0.87.

PUBLICATIONS

- [Revolutionizing Healthcare: A Review Unveiling the Transformative Power of Digital Twins](#) - IEEE Access, 2024
- [Enhancing Robustness in Audio Visual Speech Recognition: A Preprocessing Approach with Transformer and CTC Loss](#) - AMATHE 2024
- [Leukaemia: A Comparative Analysis of Deep Learning Models Using ALL Dataset](#) - RAICS 2024